Report R21 — strong or weak? Grading the fragmentation of 108/201 and 156/198

An addendum to R18 — Krylov sector analysis of quantum cellular automata

Tier 1 analysis

2026-08-14

Abstract

R9 and R19 report $D_{\max}/2^N \to 0$ for all four rules R18 is about. That is necessary for fragmentation but it is not the criterion in use: the literature grades fragmentation by the largest Krylov sector relative to the *symmetry sector* that contains it, $D_{\max}(s)/\dim s$, and dividing by 2^N instead is the same test only when the symmetry is trivial — which here it is not. We run the proper test. **All four rules are strongly fragmented, and the conclusion is not close.** At $N = 20$ the ratio in the dominant symmetry sector is 5.9×10^{-3} (W_{108}), 8.8×10^{-3} (W_{201}) and 3.0×10^{-3} (W_{156}/W_{198}), decaying geometrically at rates 0.841, 0.834 and 0.697 against the predicted $\varphi/2 = 0.809$ and $4^{1/5}/2 = 0.660$. We do not take R18's charges on trust: a detector built here finds the *complete* space of conserved charges $\sum_i q(x_i, \ldots, x_{i+r-1})$ by exact rational elimination, at ranges $r = 1, 2, 3, 4$ and on a two-site unit cell, and independently reproduces R18's pair at $r = 2$ — for W_{108} it returns literally the sublattice-resolved counts of the wall word `00`, with no notion of a wall anywhere in its input. Longer range finds *more* charges (ten for W_{108} at $r = 4$, which R18 did not report) and the verdict does not move. Underneath the numerics there is a theorem: finitely many finite-range extensive charges have at most $O(N^m)$ joint level sets, so the symmetry can only ever remove a *polynomial* factor, and any rule whose $D_{\max}$ grows like λ^N with $\lambda < 2$ is strongly fragmented for that reason alone. Finally, the diagnostic is shown not to be vacuous on this project's own rules: W_{60}/W_{102} hold exactly half the Hilbert space in one sector and grade **weak**, while W_{150} and W_{105} turn out to be **not fragmented at all** — their Krylov sectors are exactly the level sets of a single conserved charge, the total domain-wall number D for W_{150} and the staggered one S for W_{105}. For those two we give the *entire* sector distribution in closed form (§6): in bond variables both rules are the same hopping model, and the sizes are one parity class of row $N + 1$ of Pascal's triangle — $\binom{N+1}{D}$ over even D for W_{150}, and $\binom{N+1}{S+\lceil N/2\rceil}$ over $S + \lceil N/2\rceil \equiv \lceil N/2\rceil \pmod 2$ for W_{105}. Their $D_{\max}$, sector counts and frozen counts all follow, and the three separate observations R19 and R20 made about them collapse into one statement about which parity class is cut. Everything is repeated at `pbc` to $N = 18$ with the same verdict, and there the detected charge level sets reproduce R18's Tier-2 counts $(13, 18, 25$ and $12, 17, 23)$ exactly.

Contents

1 The question, stated carefully

1.1 Two ratios, one of which is the criterion

Write $\mathcal{K}(x)$ for the Krylov sector of a computational basis state x and $s(x)$ for the symmetry sector containing it. The grading used in the literature (Sala *et al.*, PRX **10**, 011047 (2020); Khemani, Hermele and Nandkishore, PRB **101**, 174204 (2020); Moudgalya, Bernevig and Regnault, Rep. Prog. Phys. **85**, 086501 (2022)) is

$$\textbf{strong:}\quad \frac{D_{\max}(s)}{\dim s} \longrightarrow 0, \qquad\qquad \textbf{weak:}\quad \frac{D_{\max}(s)}{\dim s} \longrightarrow \text{const} > 0,$$

in the thermodynamic limit, with s the dominant (largest) symmetry sector. Weak fragmentation still has one thermal Krylov sector that fills its symmetry sector, with the frozen and small sectors a vanishing remainder; strong fragmentation has no such sector, and a typical initial state explores an asymptotically vanishing fraction of what its conservation laws allow.

R9's $D_{\max}/2^N$ is the same quantity only if $\dim s = 2^N$, i.e. only if there is no symmetry to speak of. For these four rules there is: R18 certifies two independent local charges apiece. So the crude ratio understates $\dim s$ and therefore *overstates* how strongly fragmented the model is. The question is whether it overstates it enough to matter.

1.2 The trap: "all conserved quantities" makes the question empty

Taken literally, the grading is not well posed. R18 proves that the *wall-occurrence set* is conserved and is a *complete invariant*: two states share a Krylov sector if and only if the rule's minimal wall word occurs at the same positions. If one is allowed to call that a symmetry, the symmetry sectors *are* the Krylov sectors, the ratio is 1 by construction, and every fragmented model in the world is "weak".

The grading is therefore always relative to a *declared* symmetry algebra — the local charges one would write down for the model — held at fixed finite range while $N \to \infty$. The honest procedure is not to declare as few charges as possible but as *many* as exist at each range, since a larger symmetry makes $\dim s$ smaller and the ratio larger, and hence makes a "strong" verdict harder to earn. That is what this report does, and Section 2 is how.

2 Finding the charges rather than assuming them

2.1 The ansatz and the elimination

`scaling.hsf_strength` detects every charge of the form

$$Q(x) \;=\; \sum_i q\big(i \bmod p;\; s_i, s_{i+1}, \ldots, s_{i+r-1}\big),$$

with s the padded string the dynamics actually sees (`obc0` pads a frozen 0 outside each end, and those zeros are charge material for the same reason R18 found them to be wall material). The method is: label every state with its Krylov sector via the flip reduction (`graph.flip_graph`), form the count-vector difference between each state and a representative of its sector, and take the exact rational null space of the resulting row space. Every basis vector is then re-evaluated and *asserted* to be constant on every sector, so nothing rests on floating point.

Two conventions are not free choices.

The unit cell must be two sites. A brick-wall cycle is invariant under two-site translations only, so a conserved density may live on one sublattice. With $p = 1$ the detector finds exactly *one* charge for each of the four rules; with $p = 2$ it finds *two*. The second is the staggered one, and $p = 2$ is R18's "parity split" — the reason its Tier-2 detector needed one too.

Detection must be parity-resolved in N. On an open chain the two sublattices swap roles with $N \bmod 2$, so a basis found at $N = 12$ need not be conserved at $N = 13$. It happens to survive at $r = 2$, which is how an earlier pass of this module got away with transferring one basis to every N; at $r = 4$ every rule in play has at least one basis vector that breaks. The module detects separately for even and odd N, and `strength` reports `krylov_inside_sym` at every measured point — the flag that caught it, and which is asserted throughout.

2.2 It reproduces R18, from a completely different direction

Proposition 1. *At $r = 2$, $p = 2$ the detector finds exactly two independent charges for each of $W_{108}, W_{201}, W_{156}, W_{198}$, and their joint level sets coincide exactly with R18's $(|W_{\text{even}}|, |W_{\text{odd}}|)$ — the sublattice-resolved occurrence counts of the rule's minimal wall word. Verified for $N = 10, 11, 12$.*

For W_{108} the agreement is more than a coincidence of partitions: the returned integer basis is, verbatim,

$$q_1 = \big[\text{even positions: } \texttt{00} \mapsto 1\big], \qquad q_2 = \big[\text{odd positions: } \texttt{00} \mapsto 1\big],$$

i.e. $|W_{\text{even}}|$ and $|W_{\text{odd}}|$ for the word `00`, which is exactly the word R18 reports. The detector was given the sector partition and a pattern-counting ansatz and nothing else; it recovered the wall word. For the other three rules the basis is a different (equally valid) choice of coordinates on the same two-dimensional space, and the level sets agree exactly.

This is the cross-check R18 could not perform on itself: its walls and its charges both came from Tier-2 modules, whereas this detector is Tier-1 code with no wall concept in it.

2.3 What R18 did not report: longer range finds more

R18's "exactly two independent local charges" is a statement about range 2. It does not survive to range 3 or 4:

	$r=1$	$r=2$	$r=3$	$r=4$
W_{108}	0	2	6	10
W_{201}	0	2	4	8
W_{156}, W_{198}	0	2	3	7
W_{60}, W_{102}	0	0	1	2
W_{150}, W_{105}	0	1	1	1

The $r=1$ column is R18's other observation, now checked independently: there is no single-site charge for any of them, so magnetisation is not conserved — immediate once one notices the gate flips a site outright.

The growth of the other columns is what makes this section necessary rather than decorative. Ten charges could in principle carve the Hilbert space finely enough to change the verdict, and the only way to know is to measure at each range separately. Section 3 does, and Section 4 explains why the answer was never in doubt.

3 The measurements

		$D_{\max}$	charges	sectors at $N=20$					
rule	tuple	base λ	$r = 1/2/3/4$	sym.	Krylov	$D_{\max}(s^*)/\dim s^*$	decay	$\lambda/2$	verdict
W_{108}	`IIIV`	1.6180	0 / 2 / 6 / 10	66	97229	5.92e-03	0.841	0.809	strong
W_{201}	`VIII`	1.6180	0 / 2 / 4 / 8	65	55405	8.83e-03	0.834	0.809	strong
W_{156}	`IIVI`	1.3195	0 / 2 / 3 / 7	66	17711	3.02e-03	0.697	0.660	strong
W_{198}	`IVII`	1.3195	0 / 2 / 3 / 7	66	17711	3.02e-03	0.697	0.660	strong
W_{60}	`IIVV`	2.0000	0 / 0 / 1 / 2	1	21	5.00e-01	1.000	1.000	weak
W_{102}	`IVIV`	2.0000	0 / 0 / 1 / 2	1	21	5.00e-01	1.000	1.000	weak
W_{150}	`IVVI`	2.0000	0 / 1 / 1 / 1	11	11	1.00e+00	1.000	1.000	unfragmented
W_{105}	`VIIV`	2.0000	0 / 1 / 1 / 1	11	11	1.00e+00	1.000	1.000	unfragmented

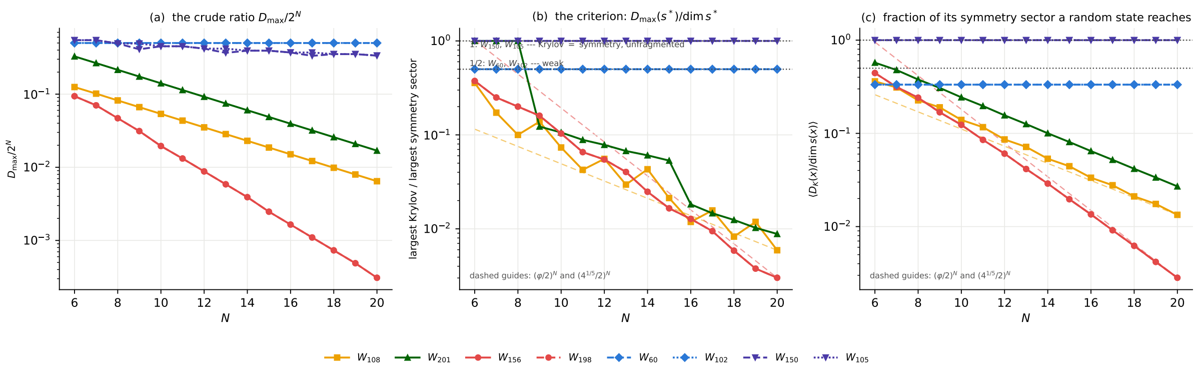

Figure 1: **The crude ratio and the criterion give the same verdict, for a reason.** (a) $D_{\max}/2^N$, as R9 reports it. (b) The criterion, $D_{\max}(s^*)/\dim s^*$ in the dominant symmetry sector at range 2; the zig-zag on the four fragmented curves is the even/odd split of the chain. (c) The average over computational basis states of $D_K(x)/\dim s(x)$ — the fraction of its own symmetry sector a random product state can reach — which is the same statement without the parity artefact. Dashed guides are $(\varphi/2)^N$ and $(4^{1/5}/2)^N$. W_{60} and W_{102} sit flat at 1/2 (weak); W_{150} and W_{105} sit flat at 1 (not fragmented at all).

The numbers behind panel (b), at range 2:

rule	N	sym. sectors	$\dim s^*$	$D_{\max}(s^*)$	$D_{\max}(s^*)/\dim s^*$	$D_{\max}/2^N$	explored
W_{108}	8	15	50	5	0.1000	0.0820	0.2262
W_{108}	12	28	612	34	0.0556	0.0352	0.0860
W_{108}	16	45	7505	89	0.0119	0.0151	0.0334
W_{108}	20	66	103008	610	0.0059	0.0065	0.0134
W_{201}	8	14	55	55	1.0000	0.2148	0.3781
W_{201}	12	27	702	55	0.0783	0.0920	0.1562
W_{201}	16	44	7899	144	0.0182	0.0394	0.0646
W_{201}	20	65	111808	987	0.0088	0.0169	0.0270
W_{156}	8	15	60	12	0.2000	0.0469	0.2415
W_{156}	12	28	658	36	0.0547	0.0088	0.0606
W_{156}	16	45	8460	108	0.0128	0.0016	0.0135
W_{156}	20	66	107404	324	0.0030	0.0003	0.0028
W_{198}	8	15	60	12	0.2000	0.0469	0.2415
W_{198}	12	28	658	36	0.0547	0.0088	0.0606
W_{198}	16	45	8460	108	0.0128	0.0016	0.0135
W_{198}	20	66	107404	324	0.0030	0.0003	0.0028

Three things to read off.

The symmetry matters, and does not save anything. At $N = 20$, W_{156}'s crude ratio is 3.1×10^{-4} while the proper one is 3.0×10^{-3} — a factor of ten. So dividing by 2^N genuinely was the wrong denominator. It is also a factor of ten and not a factor of 2^{cN}, which is why the verdict is unchanged. For W_{108} the two ratios are 6.5×10^{-3} and 5.9×10^{-3}, nearly equal.

The dominant symmetry sector holds $\Theta(2^N/N)$ states. The quantity $\dim(s^*)\,N/2^N$ sits at $1.56, 1.79, 1.83, 1.97$ for W_{108} over $N = 8, 12, 16, 20$ and similarly for the others — flat to within a factor, which is the numerical content of "the symmetry only removes a polynomial factor". The number of symmetry sectors itself runs $15, 28, 45, 66$ at $N = 8, 12, 16, 20$, i.e. $\binom{N/2+2}{2}$, visibly quadratic.

The decay rate is the predicted one. With no charges at all ($r = 1$) the measured rate is 0.8090 for W_{108}, against $\varphi/2 = 0.809017$, and 0.6581 for W_{156} against $4^{1/5}/2 = 0.659754$ — six and three digits respectively, with the exponent derived (R18's wall recurrence, R19's closed forms, R20's OEIS entries) rather than fitted. Switching the charges on lifts the rate slightly, to 0.841 and 0.697, which is exactly the $O(N^2)$ prefactor showing up as a slower apparent decay over a finite window.

3.1 Every range, separately

rule	range r	charges	sym. sectors	$D_{\max}(s^*)/\dim s^*$	decay	verdict
W_{108}	1	0	1	0.0065	0.809	strong
W_{108}	2	2	66	0.0059	0.841	strong
W_{108}	3	6	1208	0.0289	0.687	strong
W_{108}	4	10	4411	0.0289	0.687	strong
W_{201}	1	0	1	0.0169	0.809	strong
W_{201}	2	2	65	0.0088	0.834	strong
W_{201}	3	4	382	0.0229	0.660	strong
W_{201}	4	8	2405	0.0241	0.665	strong
W_{156}	1	0	1	0.0003	0.658	strong
W_{156}	2	2	66	0.0030	0.697	strong
W_{156}	3	3	111	0.0055	0.760	strong
W_{156}	4	7	972	0.0145	0.717	strong
W_{198}	1	0	1	0.0003	0.658	strong
W_{198}	2	2	66	0.0030	0.697	strong
W_{198}	3	3	111	0.0055	0.760	strong
W_{198}	4	7	972	0.0145	0.717	strong
W_{60}	1	0	1	0.5000	1.000	weak
W_{60}	2	0	1	0.5000	1.000	weak
W_{60}	3	1	2	1.0000	1.000	weak
W_{60}	4	2	3	1.0000	1.000	weak
W_{102}	1	0	1	0.5000	1.000	weak
W_{102}	2	0	1	0.5000	1.000	weak
W_{102}	3	1	2	0.5000	1.000	weak
W_{102}	4	2	3	1.0000	1.000	weak
W_{150}	1	0	1	0.3364	0.976	marginal (polynomial)
W_{150}	2	1	11	1.0000	1.000	unfragmented
W_{150}	3	1	11	1.0000	1.000	unfragmented
W_{150}	4	1	11	1.0000	1.000	unfragmented
W_{105}	1	0	1	0.3364	0.976	marginal (polynomial)
W_{105}	2	1	11	1.0000	1.000	unfragmented
W_{105}	3	1	11	1.0000	1.000	unfragmented
W_{105}	4	1	11	1.0000	1.000	unfragmented

All four rules grade **strong** at $r = 1, 2, 3$ and 4. One caveat belongs with the $r = 4$ rows and is stated rather than buried: with $m = 10$ charges the bound below carries an N^{10} prefactor, and $N \leq 20$ is nowhere near the asymptotic regime for such a prefactor. The $r = 4$ ratios are therefore noisy and occasionally sit at 1 at small N. They are reported because suppressing them would be worse; the argument that settles $r = 4$ is the theorem, not the fit.

4 Why the answer could not have been otherwise

Theorem 1. *Fix a range r and unit cell p, and let $Q_1, \ldots, Q_m$ be conserved charges of the above form, $m \leq p\,2^r$. Then the number of joint level sets is $O(N^m)$, and consequently*

$$\frac{D_{\max}(s^*)}{\dim s^*} = O\left(N^m \frac{D_{\max}}{2^N}\right).$$

If $D_{\max} = \Theta(\lambda^N)$ with $\lambda < 2$, the ratio vanishes exponentially, at rate $\lambda/2$ up to the polynomial factor, and the fragmentation is strong.

Proof. Each Q_j is a sum of at most $L = N + 2$ terms drawn from the finite set of values of q_j, so Q_j takes $O(N)$ distinct values and the m of them have $O(N^m)$ joint level sets. The largest of those level sets therefore holds at least $2^N/O(N^m)$ states, and $D_{\max}(s^*) \leq D_{\max}$. $\square$

Three consequences worth stating plainly.

1. **The user's instinct was right, and the reason is this theorem.** "$D_{\max}/2^N \to 0$, so it should be strong" is not a sloppy shortcut when the decay is exponential: a finite-range symmetry cannot supply more than a polynomial denominator, so the two criteria agree. What the theorem also shows is where the shortcut fails — when the decay is only polynomial, the symmetry can absorb all of it, and the two criteria come apart. Both cases occur in this project (Section 5).

2. **Range cannot rescue the symmetry story, at any fixed r.** Ten charges give N^{10}, which is enormous at $N = 20$ and irrelevant at $N = 200$. The only way a symmetry could compete is if m grew with N — i.e. if the declared symmetry were not of fixed finite range — and that is precisely the degenerate case of Section 1.2, in which the wall set itself is the symmetry.

3. **It sharpens what R18 proved.** R18 showed the charges are too *coarse* to explain the sectors: $O(N^2)$ level sets against ρ^{2N} or φ^N Krylov sectors, the ratio growing exponentially. That is a statement about counting sectors. The theorem here is the corresponding statement about their *sizes*, and it is what turns "the symmetry is not the explanation" into "the fragmentation is strong".

5 The diagnostic is not vacuous: three grades among eight rules

A test that returns "strong" for everything would say nothing. Running the same pipeline over the polynomial corner of the unitary sweep produces three distinct outcomes, all within this project's own rules.

W_{60}, W_{102} — weak. $D_{\max} = 2^{N-1}$: half the Hilbert space sits in one Krylov sector, at every N. There is no charge at all below range 3, so the symmetry sector is the whole space and $D_{\max}(s^*)/\dim s^* = 1/2$ exactly, forever. The average explored fraction is exactly 1/3. This is the textbook weak picture — one thermal sector filling its symmetry sector, with N small sectors beside it — realised by a rule of this project rather than a model imported for the comparison.

W_{150}, W_{105} — not fragmented at all. Both have exactly *one* conserved range-2 charge, and its level sets *are* their Krylov sectors: $n_{\text{sym}} = n_{\text{Krylov}} = 11$ at $N = 20$, ratio identically 1, explored fraction identically 1, at every N and every range. Calling this "weak" would be wrong; weak fragmentation still has small sectors beside the thermal one, and these rules have none. The charges are named:

Proposition 2. *W_{150} conserves the total domain-wall number $D = \sum_i b_i$ and not the staggered S; W_{105} conserves $S = \sum_i (-1)^i b_i$ and not D. In each case the level sets of that single charge are exactly the Krylov sectors.*

The mechanism, and the full sector distribution that follows from it, are in §6: W_{105}'s move creates and destroys walls in *pairs*, which is why D moves, and the pairing is staggered, which is why S does not.
This retro-explains two things. It explains why W_{150}'s sector count is the trivial $\lfloor (N+1)/2 \rfloor + 1$: the `obc0` padding forces an even number of domain walls among the $N + 1$ bonds, so D takes exactly that many values — connecting to R8's domain-wall count for rule 150, and matching R20's identification of the count as `A008619`$(N + 1)$. And it explains R20's identifications of their $D_{\max}$: W_{105}'s largest sector is the central binomial coefficient $\binom{N+1}{\lfloor (N+1)/2 \rfloor}$ because that is the largest level set of a balanced counting statistic, and W_{150}'s is `A214282` for the same reason. Those closed forms are the symmetry showing through, not accidents.

W_{108}, W_{201}, W_{156}, W_{198} — **strong.** The four rules of R18, at every range tested, by both the finite-N measurement and the theorem.

The interesting structural fact is that the split is exactly the split in λ: the strong rules are the ones with $\lambda = \varphi$ or $4^{1/5}$, both < 2; the weak and unfragmented ones all have $\lambda = 2$, differing only in the sub-exponential correction (2^{N-1} against $2^N/\sqrt{N}$). By the theorem that is not a coincidence but the whole content of the classification.

6 W_{150} and W_{105} in closed form

Section 5 states that each of these two rules has a single conserved charge whose level sets are its Krylov sectors, and names the charges. That is enough to grade them, but it leaves the sector *distribution* implicit. It is worth writing down, because both rules turn out to be the same model in different variables, and the whole distribution is one row of Pascal's triangle split by parity.

6.1 Bond variables make them the same model

Let $s = (0, x_1, \ldots, x_N, 0)$ be the padded string and

$$u_i \;=\; s_i \oplus s_{i+1}, \qquad i = 0, \ldots, N$$

its $N+1$ bond variables. Because `obc0` pins both ends to 0, the map $x \mapsto u$ is a *bijection* from the 2^N states onto the **even-weight** vectors of $\{0,1\}^{N+1}$. Flipping site i toggles exactly u_{i-1} and u_i, and the two rules are complementary in when they fire:

$$W_{150} = \texttt{IVVI}: \text{ fire iff } x_{i-1} \neq x_{i+1}, \qquad W_{105} = \texttt{VIIV}: \text{ fire iff } x_{i-1} = x_{i+1},$$

which in bond variables reads $u_{i-1} \neq u_i$ and $u_{i-1} = u_i$ respectively. So

- W_{150} toggles the pair (u_{i-1}, u_i) exactly when they *differ*: $01 \leftrightarrow 10$. A wall **hops**, and $D = \text{weight}(u)$ is conserved.
- W_{105} toggles the pair exactly when they *agree*: $00 \leftrightarrow 11$. A wall **pair is created or destroyed**, so D is *not* conserved — confirming the asymmetry directly. But in the *staggered* bond variable
$$v_i \;=\; u_i \oplus (i \bmod 2)$$
the condition $u_{i-1} = u_i$ becomes $v_{i-1} \neq v_i$, so in v the rule is the identical hopping model and weight(v) is conserved.

Both rules are therefore adjacent transpositions on a chain of $N+1$ bonds, which is why in each case the level sets of the single charge *are* the Krylov sectors: the hops generate the full symmetric group on the positions, so a fixed weight is one orbit. The only thing that distinguishes them is which residue class of weights the boundary constraint allows. Since v differs from u at the $\lceil N/2 \rceil$ odd positions,

$$\text{weight}(u) \text{ even} \quad\Longleftrightarrow\quad \text{weight}(v) \equiv \lceil N/2 \rceil \pmod 2.$$

6.2 The distributions

Writing the charges as R18 does — $D = \sum_i b_i$ and $S = \sum_i (-1)^i b_i$, with $b_i = u_i$ — one has $D = \text{weight}(u)$ and $S = \text{weight}(v) - \lceil N/2 \rceil$, and:

Proposition 3 (sector distribution at `obc0`).

$$\boxed{W_{150}: \quad \left|\,\text{sector } D\,\right| \;=\; \binom{N+1}{D}, \qquad D = 0, 2, 4, \ldots \le N+1}$$

$$\boxed{W_{105}: \quad \left|\,\text{sector } S\,\right| \;=\; \binom{N+1}{S + \lceil N/2 \rceil}, \qquad S + \lceil N/2 \rceil = \varepsilon, \varepsilon+2, \ldots \le N+1}$$

with $\varepsilon = \lceil N/2 \rceil \bmod 2$. *In both cases the label is even — D because the padded chain has an even number of walls, S because $S = -2\sum_i (-1)^i x_i x_{i+1}$. Verified against the engine for $N = 3 \ldots 18$: sizes, labels, sector count, and the charge being constant on each Krylov sector.*

So the answer to "for which w is $\binom{N+1}{w}$ a sector size": w **even for** W_{150}**;** $w \equiv \lceil N/2 \rceil \pmod 2$ **for** W_{105}. Both distributions are one parity class of row $N+1$, and $\sum_w \binom{N+1}{w} = 2^N$ over either class, as it must be.

rule	N	label	values	sector sizes
W_{150}	8	D	0, 2, 4, 6, 8	1, 36, 126, 84, 9
W_{105}	8	S	-4, -2, 0, 2, 4	1, 36, 126, 84, 9
W_{150}	9	D	0, 2, 4, 6, 8, 10	1, 45, 210, 210, 45, 1
W_{105}	9	S	-4, -2, 0, 2, 4	10, 120, 252, 120, 10
W_{150}	10	D	0, 2, 4, 6, 8, 10	1, 55, 330, 462, 165, 11
W_{105}	10	S	-4, -2, 0, 2, 4, 6	11, 165, 462, 330, 55, 1
W_{150}	11	D	0, 2, 4, 6, 8, 10, 12	1, 66, 495, 924, 495, 66, 1
W_{105}	11	S	-6, -4, -2, 0, 2, 4, 6	1, 66, 495, 924, 495, 66, 1

6.3 Everything else follows

Sector counts. W_{150} has $\lfloor (N+1)/2 \rfloor + 1$ sectors (R20's `A008619`$(N+1)$); W_{105} has $\lfloor (N+1-\varepsilon)/2 \rfloor + 1$, the period-4 sequence $\ldots, 5, 5, 6, 7, 7, 7, 8, \ldots$ which is `A004525`$(N+2)$ — an identification R20 was missing and which the closed form makes obvious what to look for.

$D_{\max}$**, and why R20 saw the two rules diverge.** W_{105} always reaches the *central* binomial, because $\lfloor (N+1)/2 \rfloor = \lceil N/2 \rceil$ is precisely the class its weights live in — so $D_{\max}(W_{105}) = \binom{N+1}{\lfloor (N+1)/2 \rfloor}$, which is R20's `A001405`$(N+1)$, now derived rather than matched. W_{150} reaches it too whenever $N+1$ is odd (the peak value then sits at two indices of opposite parity, so both classes get it) or $N+1 \equiv 0 \pmod 4$. It misses it exactly at $N \equiv 1 \pmod 4$, which is why R20 observed the two parting company at $N = 9$ (210 against 252) and $N = 13$ (3003 against 3432) — and $17, 21, \ldots$ likewise.

Frozen states. A sector has size 1 iff its label is at an end of the row, $w = 0$ or $w = N+1$. For W_{150}, $w = 0$ is always allowed and $w = N+1$ iff N is odd: $n_{\text{froz}} = 1 + (N \bmod 2)$, R19's period-2 law. For W_{105} the class admits $w = 0$ iff $\varepsilon = 0$ and $w = N+1$ iff $N+1 \equiv \varepsilon$, giving $1, 0, 1, 2$ for $N \equiv 0, 1, 2, 3 \pmod 4$ — R19's period-4 law, including the zero that had to be special-cased in `paper_figures`.

One fact that ties the residues together. The reflection $w \mapsto N+1-w$ is a symmetry of row $N+1$, and it swaps the two parity classes exactly when $N+1$ is odd. Hence W_{150} and W_{105} have the *same multiset of sector sizes* for every N **except** $N \equiv 1 \pmod 4$. That is the same residue at which W_{105} has no frozen states at all, and the same residue at which their $D_{\max}$ differ: at $N \equiv 1 \pmod 4$ the allowed class excludes both ends of the row and contains the centre, and at every other N it does the opposite or coincides. The three observations R19 and R20 recorded separately are one statement about which parity class row $N+1$ is cut into.

7 The same question on the ring

R18 tested both boundary conditions, so this addendum does too. The pipeline is exhaustive enumeration to $N = 18$ — the same kind of computation R18 already ran at `pbc`, not the production sweep.

		$D_{\max}$	charges	sectors at $N = 18$					
rule	tuple	base λ	$r = 1/2/3/4$	sym.	Krylov	$D_{\max}(s^*)/\dim s^*$	decay	$\lambda/2$	verdict
W_{108}	`IIIV`	1.6180	0 / 2 / 5 / 9	50	24914	7.58e-03	0.681	0.809	strong
W_{201}	`VIII`	1.6180	0 / 2 / 5 / 9	50	24914	7.58e-03	0.681	0.809	strong
W_{156}	`IIVI`	1.3195	0 / 2 / 3 / 6	47	5779	4.73e-03	0.683	0.660	strong
W_{198}	`IVII`	1.3195	0 / 2 / 3 / 6	47	5779	4.73e-03	0.683	0.660	strong
W_{60}	`IIVV`	2.0000	0 / 0 / 0 / 0	1	2	1.00e+00	1.000	1.000	weak
W_{102}	`IVIV`	2.0000	0 / 0 / 0 / 0	1	2	1.00e+00	1.000	1.000	weak
W_{150}	`IVVI`	2.0000	0 / 1 / 1 / 1	10	12	1.00e+00	1.000	1.000	weak
W_{105}	`VIIV`	2.0000	0 / 1 / 1 / 1	9	9	1.00e+00	1.000	1.000	unfragmented

The verdict is the same: **strong for all four rules at every range**, with $D_{\max}(s^*)/\dim s^*$ at 7.6×10^{-3} (W_{108}, W_{201}) and 4.7×10^{-3} (W_{156}, W_{198}) at $N = 18$. Three details are specific to the ring.

A third confirmation of R18, on the nose. At range 2 the detected charges have $13, 18, 25$ joint level sets for W_{108} at $N = 8, 10, 12$ and $12, 17, 23$ for W_{156} — exactly the numbers R18 quotes for Tier-2's *full certified charge set* at `pbc`. That was R18's own check that nothing conserved had been left out; it now reproduces from a third, independent detector.

W_{108} and W_{201} merge. On the ring their series coincide term for term (same charge counts, same sector counts, same ratios), because the spin flip that relates them is a `pbc` symmetry and not an `obc0` one — consistent with R19's finding that the `obc0` vacuum breaks it.

The controls move, and in the expected direction. W_{60} and W_{102} lose their charges entirely (none at any range up to 4) and collapse to *two* Krylov sectors of 2^{N-1} each, so the ratio is 1 rather than 1/2; the periodic chain removes the boundary that split the smaller sectors off. W_{105} stays unfragmented (9 sectors, 9 level sets at $N = 18$), while W_{150} acquires a small excess (12 Krylov sectors inside 10 level sets) and grades weak rather than unfragmented. None of this touches the four rules of R18.

8 Answering the question as asked

1. **Does $D_{\max}/2^N \to 0$ imply strong HSF?** Not in general — but yes whenever the decay is exponential and the declared symmetry is of fixed finite range, which covers all four rules here. The two criteria come apart only in the polynomial case, and this project contains examples of that too.

2. **Do 108, 201, 156, 198 exhibit weak or strong HSF, measured against the R18 charges? Strong**, at $N = 20$, at ranges 1 through 4, with the ratio at 5.9×10^{-3}, 8.8×10^{-3}, 3.0×10^{-3} and 3.0×10^{-3} and decaying geometrically at the predicted rates.

3. **Were S, D, n_{01}, n_{10} the right charges to divide by?** They are a spanning set but not an independent one: for each rule the independent pair at range 2 is the sublattice-resolved wall counts, of which D or S is a function (R18, confirmed here by an independent detector). Dividing by fewer would have given a larger $\dim s$ and an even smaller ratio,

so the choice made here is the conservative one. Dividing by *more* — the range-3 and range-4 charges R18 did not report — also leaves the verdict unchanged.

4. **Is there anything the symmetry could have done?** No, and the theorem says why: m finite-range charges have $O(N^m)$ level sets, so no choice of local symmetry can supply an exponential denominator. The only way to close the gap is with the wall set itself, which is not a local charge and is the entire point of R18.

9 Reproduce

```
python -m qca_fragmentation.scaling.hsf_strength \
       --bc obc0 --rebuild --n-max 20
python -m qca_fragmentation.scaling.hsf_strength \
       --bc pbc  --rebuild --n-max 18
python -m pytest qca_fragmentation/tests/test_hsf_strength.py -q
```

Everything here comes from `analytics/hsf_strength_{bc}.json`. The Krylov labels use the flip reduction of `graph.flip_graph`, valid because all eight rules are unitary; the charge detector, the ratios and the figure are `scaling/hsf_strength.py`, which depends on R18's `scaling/wall_charges.py` only for the cross-check in §2.2 and for naming D and S in §5.